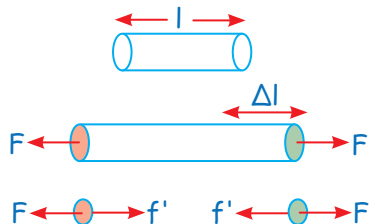


Yeh ek bahut chota sa mazedaar chapter hai. Jisse kabhi-kabhi directly aur kabhi-kabhi mechanics, wave, thermal expansion se mix karke question pooche jaate hain. Iss chapter mein kuch formula aayenge, woh galti se bhi nahi bhoolna hai



ELASTICITY

Property due which body regains its original shape & size after removal of deforming forces.



$$\Rightarrow F = f'$$

Here, f' = restoring force developed.

SKC

Khichne ki nhi, Regain krne ki property ko Elasticity Kehte hai
(steel is more elastic then rubber)

Stress

$$\text{Stress} = \text{Restoring force per unit area} = \frac{f'}{A} = \frac{F}{A}$$

Is case mein restoring force surface ke normal develop hoti hai, isliye stress ko normal stress ya longitudinal stress bhi kehte hain.

Strain

$$\text{Strain} = \frac{\Delta l}{l} = \frac{\text{Change in length}}{\text{Original length}}$$

HOOK'S LAW

Acc. to hook's law within proportional limit

$$\text{Stress} \propto \text{strain}$$

$$\text{Stress} = \gamma(\text{strain})$$

★ γ = Young Modulus of elasticity depend on material.

$\gamma \uparrow \Rightarrow \text{elasticity} \uparrow \text{ more elastic}$

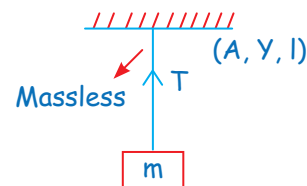


$$\frac{F}{A} = \gamma \left(\frac{\Delta l}{l} \right) \Rightarrow \Delta l = \frac{Fl}{AY}$$

Area of cross section $A = \pi r^2$
Iss chapter mein sabse zyada important formula har saal iss par question aate hain



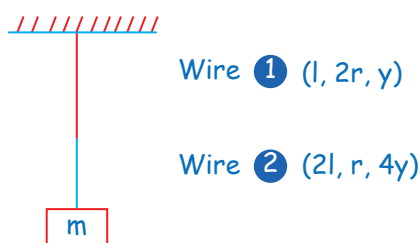
Q. A block of mass m is suspended by a elastic massless wire (A, Y, l) find elongation in wire.



$$\text{Sol. } \frac{F}{A} = \gamma \frac{\Delta l}{l} \Rightarrow \frac{\text{Tension}}{A} = \gamma \frac{\Delta l}{l}$$

$$\frac{mg}{A} = \gamma \frac{\Delta l}{l} \Rightarrow \Delta l = \frac{mgl}{Ay} = \frac{mgl}{\pi r^2 \gamma}$$

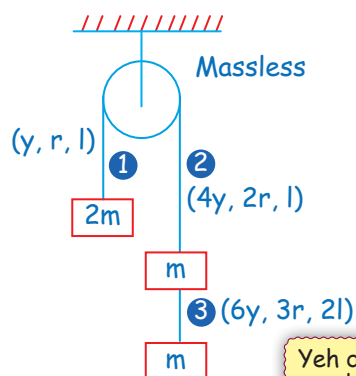
Q. Find ratio of elongation in both the wire in following figure if wire are massless.



Sol. $\Delta l_1 = \frac{fl}{Ay} = \frac{fl}{\pi 4r^2 y}$

$$\frac{\Delta l_1}{\Delta l_2} = \frac{fl}{\pi 4r^2 y} \times \frac{\pi r^2 4y}{2fl} = \frac{1}{2}$$

Q. Find the elongation in each wire.



Sol. $\Delta l_1 = \frac{fl}{Ay} = \frac{fl}{\pi r^2 y} = \frac{2mgl}{\pi r^2 y}$

$$\Delta l_2 = \frac{fl}{Ay} = \frac{fl}{\pi 4r^2 4y} = \frac{2mgl}{\pi 4r^2 4y}$$

$$\Delta l_3 = \frac{fl}{Ay} = \frac{2fl}{\pi 9r^2 6y} = \frac{2mgl}{\pi 9r^2 6y}$$

Yeh dhyaan se dekhna hai ki kis wire mein kitni tension developed hui hai.



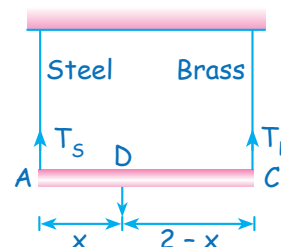
Q. A light rod of length 2.00 m is suspended from the ceiling horizontally by means of two vertical wires of equal length tied to its ends. One of the wires is made of steel and is of cross-section 10^{-3} m^2 and the other is of brass of cross-section $2 \times 10^{-3} \text{ m}^2$. Find out the position along the rod at which a weight may be hung to produce,

(a) equal stresses in both wires

(b) equal strains on both wires.

Young's modulus for steel is $2 \times 10^{11} \text{ N/m}^2$ and for brass is 10^{11} N/m^2 .

Sol. (a) Given, stress in steel = stress in brass



$$\therefore \frac{T_s}{A_s} = \frac{T_b}{A_b}$$

$$\therefore \frac{T_s}{T_b} = \frac{A_s}{A_b} = \frac{10^{-3}}{2 \times 10^{-3}} = \frac{1}{2} \quad \dots (i)$$

As the system is in equilibrium, taking moments about D, we have

$$T_s \cdot x = T_b (2 - x)$$

$$\therefore \frac{T_s}{T_b} = \frac{2-x}{x} \quad \dots (ii)$$

From Eqs. (i) and (ii), we get

$$x = 1.33 \text{ m}$$

(b) Strain = $\frac{\text{Stress}}{y}$

Given, strain in steel = strain in brass

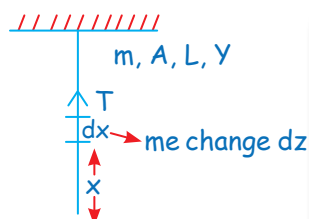
$$\therefore \frac{T_s / A_s}{y_s} = \frac{T_b / A_b}{y_b}$$

$$\therefore \frac{T_s}{T_b} = \frac{A_s y_s}{A_b y_b} = \frac{(1 \times 10^{-3})(2 \times 10^{11})}{(2 \times 10^{-3})(10^{11})} = 1 \quad \dots (iii)$$

From Eqs. (ii) and (iii), we have

$$x = 1.0 \text{ m}$$

Q. Find the elongation in rod due to its own weight.



Sol. $L \rightarrow m$

$$1 \rightarrow m/L$$

$$x \rightarrow \frac{m}{L} x$$

$$\frac{f}{A} = Y \text{ (Strain)}$$

$$\frac{m \times g}{L} = Y \frac{dz}{dx}$$

Dhyaan se dekho yeh mass wali rassi hai



Jismein alag-alag jagah alag-alag tension hai. Toh alag-alag element ka elongation alag-alag hoga. Hence humne ex ko yaad kiya, bole toh x par jaake dx element pakda jismein let dz elongation developed hua

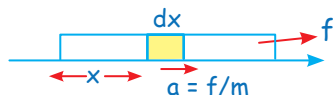
$$\frac{mg}{LAY} \int_0^L x dx = \int dz = \text{Total elongate}$$

$$(\Delta l)_{\text{net}} = \frac{mg}{LAY} \frac{L^2}{2} = \frac{mgL}{2AY}$$

Q. A rod is being pulled by a force f as shown in the figure. Find elongation in rod.



Sol.



$$T = \left(\frac{m}{L}\right) x \cdot a = \frac{mx}{L} \frac{f}{M} = \frac{fx}{L}$$

$$\text{Stress} = Y(\text{Strain})$$

$$\frac{T}{A} = Y \frac{dz}{dx}$$

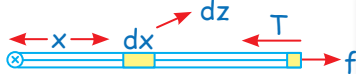
$$\frac{fx}{LA} = Y \frac{dz}{dx}$$

$$\int dz = \int_0^L \frac{f}{LAY} x dx$$

$$\Delta l_{\text{net}} = \frac{f}{LAY} \cdot \frac{L^2}{2} = \frac{fL}{2AY}$$

Q. fix m, L, Y, A $\vec{a} = 0$ find elongation in rod.

Sol.



$$\frac{T}{A} = Y \frac{dz}{dx} \Rightarrow \int_0^L \frac{f}{AY} dx = \int dz$$

$$\Delta l = \frac{fL}{AY}$$

$$\frac{f}{A} = \frac{Y \Delta l}{L} \quad (T = f) \quad [\text{Direct}]$$

Abe yahan har jagah tension same hai, isliye bina integration se direct answer nikaal sakte ho.



Q. (m, A, L, Y)

→ Find total elongation in rod,

Sol.



$$T \leftarrow \text{rod} \rightarrow T + dT$$

$$a_c = x\omega^2$$

$$T - (T + dT) = (dm) \times \omega^2$$

$$-dT = \frac{m}{L} dx \cdot x \omega^2$$

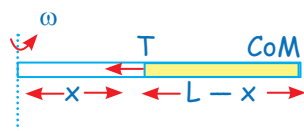
$$dm = \lambda dx = \frac{m}{L} dx$$

$$\int_0^T dT = \frac{-m\omega^2}{L} \int_L^x x dx$$

$$T = \frac{-m}{L} \omega^2 \left(\frac{x^2 - L^2}{2} \right)$$

$$T = \frac{m\omega^2}{2L} (L^2 - x^2)$$

$$\text{Stress} = \frac{T}{A} = \frac{m\omega^2}{2LA} (L^2 - x^2)$$



Iska ek aur mast wala method dekh



$$\vec{F}_{\text{net}} = m_{\text{total}} \vec{a}_{\text{com}}$$

$$T = \frac{m}{L} (L - x) r \omega^2$$

$$T = \frac{m}{L} (L - x) \left(x + \frac{L - x}{2} \right) \omega^2$$

$$T = \frac{m}{L} (L - x) \left(\frac{L + x}{2} \right) \omega^2$$

$$T = \frac{m}{L} \frac{\omega^2}{2} (L^2 - x^2)$$

$$\star F = \frac{YA}{L} \Delta L \text{ and } F = K \Delta L$$

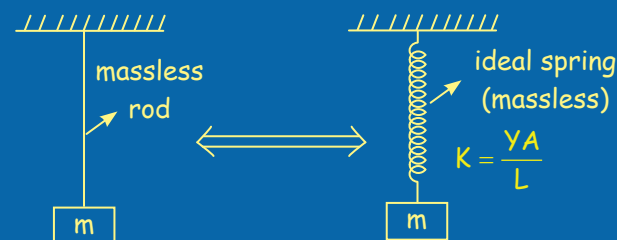
Where K is spring constant.

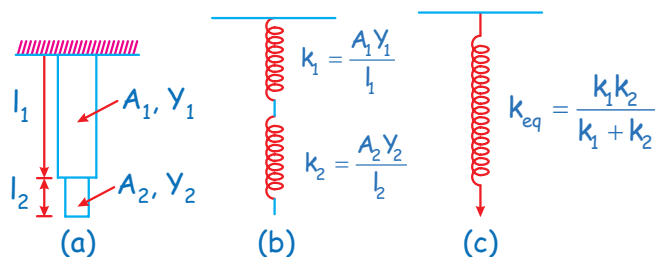
$$\text{So, } K \Delta L = \frac{YA}{L} \Delta L \Rightarrow K = \frac{YA}{L}$$

SKC

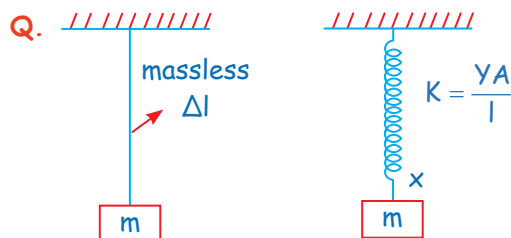


Ek elastic massless rod ko mai ek aese ideal spring se replace kar sakta hoon jiska spring constant





ELASTIC POTENTIAL ENERGY



Sol. Its elongation is x

$$P.E = \frac{1}{2} Kx^2$$

$$P.E = \frac{1}{2} \frac{YA}{l} \left(\frac{\Delta l}{l} \right)^2 x l^2$$

$$P.E = \frac{1}{2} Y \left(\frac{Al}{Vol^n} \right) \left(\frac{\Delta l}{l} \right)^2$$

$$\frac{P.E}{Vol^n} = \frac{1}{2} Y (\text{Strain})^2$$

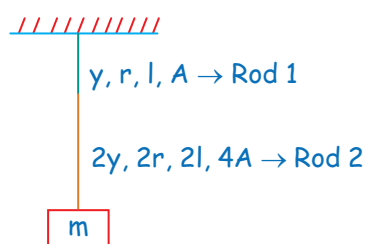
$$= \frac{1}{2} \frac{\text{Stress}}{\text{Strain}} \times (\text{Strain})^2 = \frac{1}{2} (\text{Stress})(\text{Strain})$$

$$= \frac{1}{2} (\text{Stress}) \left(\frac{\text{Stress}}{Y} \right) = \frac{1}{2} \frac{\text{Stress}^2}{Y}$$

$$\frac{P.E}{Vol^n} = \frac{1}{2} \frac{(\text{Stress})^2}{Y} = \frac{1}{2} Y (\text{Strain})^2$$

yah formula galti se bhi mt bhulna

Q. Find ratio of (PE) stored in rod



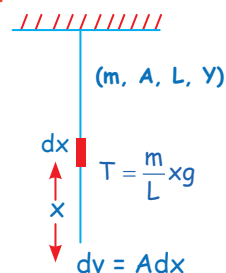
Sol. $\frac{P.E}{Vol^n} = \frac{1}{2} \frac{(\text{Stress})^2}{Y}$

$$(PE)_{\pm} = \frac{1}{2} \left(\frac{f}{A} \right)^2 \frac{(Vol^n)}{Y} = \frac{1}{2} \frac{T^2}{A^2} \frac{Al}{Y}$$

$$P.E = \frac{T^2}{2} \cdot \frac{l}{AY}$$

$$\frac{(P.E)_1}{(P.E)_2} = \frac{\frac{l_1}{l_2} \frac{A_2}{A_1} \frac{Y_2}{Y_1}}{1} = \frac{1}{2} \times 4 \times 2 = 4$$

Q. Find total elastic P.E stored in rod due to its own weight.



Sol. $\frac{d(PE)}{dv} = \frac{(\text{Stress})^2}{2Y} = \frac{1}{2Y} \left(\frac{T}{A} \right)^2$

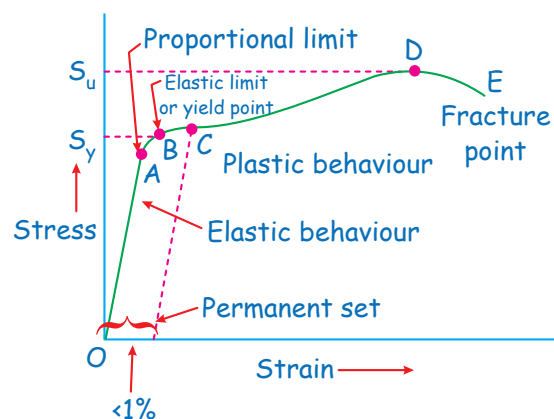
$$= \frac{1}{2YA^2} \left(\frac{m}{L} xg \right)^2 = \frac{m^2 g^2}{2YA^2 L^2} x^2$$

$$\int d(PE) = \int_0^L \frac{1}{2} \frac{m^2 g^2 x^2}{YA^2 L^2} (Adx)$$

$$P.E = \frac{1}{6} \frac{m^2 g^2 L}{Ay}$$

STRESS-STRAIN CURVE

Ab hum stress vs strain ka graph dekhenge, jo ki ek experimental graph hai aur alag-alag material ke liye iski shakl alag-alag hoti hai.



The region between O to A, the curve is linear. In this region, Hook's law is obeyed. The body regains its original dimensions when the applied force is removed. In this region, the solid behaves as an **elastic body**. In the region from A to B, stress and strain are **not proportional**. But, the body still returns to its original dimension when the load is removed.

The point B in the curve is known as **yield point** (also known as elastic limit) and the corresponding stress is known as **yield strength** (S_y) of the material.

If the tensile or compressive stress exceeds the proportional limit, the strain is no longer proportional to the stress (see the figure).

If the stress exceeds the elastic limit, the material is permanently deformed.

The maximum stress that can be withstood without breaking is called the **ultimate strength**.

A ductile material continues to stretch beyond its ultimate tensile strength without breaking.

O → A

- ★ St. line
- ★ $y = mx$
- ★ Stress $\propto$ Strain
- ★ Stress = Y Strain
- ★ Slope = $\tan\theta = y$
- ★ hook's law valid
- ★ The body returns to its original dimension when the load is removed.

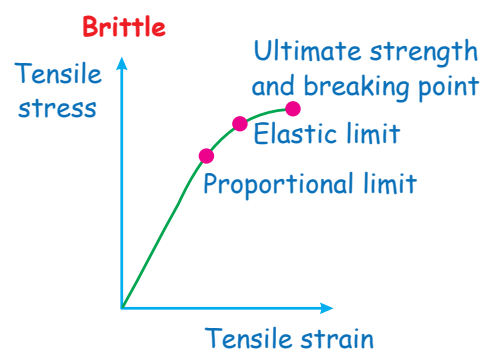
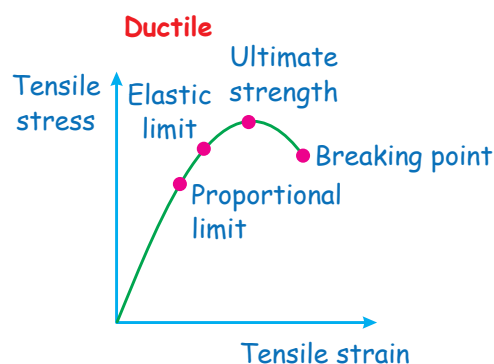
A → B

- ★ Non linear
- ★ Hook's law not valid
- ★ The body still returns to its original dimension when the load is removed.

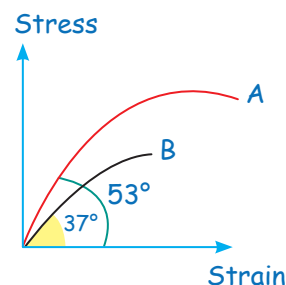
★ **Proportional Limit:** The limit in which Hook's law is valid and stress is in directly proportional to strain is called proportional limit Stress $\propto$ Strain.

★ **Elastic Limit:** That maximum stress which on removing the deforming force makes the body to recover completely its original state.

★ **Breaking Point:** The position when the strain becomes so large that the wire breaks down at last, is called breaking point. At this position the stress acting in that wire is called breaking stress and strain is called Breaking strain.



Q. Which material (A or B) is more elastic?



Sol. $\frac{Y_A}{Y_B} = \frac{\tan 53^\circ}{\tan 37^\circ}$

$\tan\theta \uparrow \rightarrow Y \uparrow$ elastic $\uparrow$

$Y_A > Y_B$

Q. A light wire of length 4m is suspended to the ceiling by one of its ends. If its cross-sectional area is 19.6 mm^2 , what is its extension under a load of 10kg. Young's modulus of steel = $2 \times 10^{11} \text{ Pa}$.

Sol. Given; original length $L = 4\text{m}$;

force $F = 10 \times 9.8 = 98 \text{ N}$;

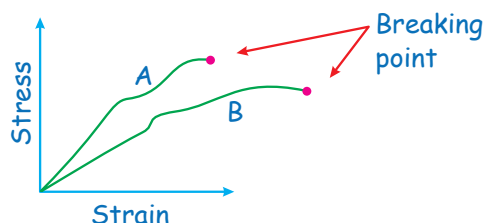
and $Y = 2 \times 10^{11} \text{ Nm}^{-2}$; $\Delta L = ?$

We have the relation,

$\Delta L = \frac{FL}{YA}$

$\therefore \Delta L = \frac{98 \times 4}{2 \times 10^{11} \times 19.6 \times 10^{-6}} = 1 \times 10^{-4} \text{ m}$
 $= 0.1 \text{ mm}$

Q. Which material (A or B) is more ductile?



Sol. $Y = \frac{\text{stress}}{\text{strain}}$ for same strain (stress A) > (stress B)

$$Y_A > Y_B$$

Strain required for breaking point of B > Strain required for breaking point of A.

So, A is less ductile.

Stress

- Longitudinal Stress $\left(\frac{F_{\perp}}{A}\right)$
- Tangential Stress/Shear Stress $(F_{\text{tangential}}/A)$
- Bulk Stress/Volⁿ Stress $\left(\frac{dF}{dA} = P\right)$

Strain

- Longitudinal strain = $\Delta l/l$
- Tangential strain = $\Delta x/x$
- Volⁿ strain = $\Delta v/v$

HOOK'S LAW

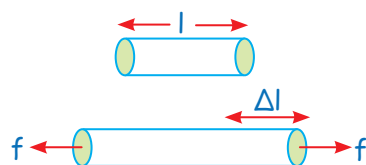
Within elastic limit

Stress $\propto$ Strain

Stress = E Strain

↳ Modulus of elasticity (y)

Longitudinal Stress: Restoring force $\perp^{\text{ar}}$ surface area. (Normal Stress)



$$\text{Stress} = \frac{F}{A} = \text{Longitudinal Stress}$$

$$\text{Longitudinal Strain} = \Delta l/l$$

$$E = Y = \frac{\text{Stress}}{\text{Strain}} = \frac{F/A}{\Delta l/l}$$

Tensile Stress:



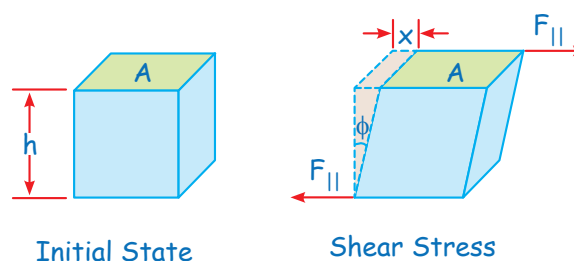
Compressive Stress:

$E = Y \rightarrow$ Young's Modulus

$$\text{Stress} = Y \text{ Strain}$$



Shearing Stress (Tangential stress) Restoring force develops in tangent to the surface area (Shear Stress)



$$\text{Tangential Stress} = \frac{F}{A}$$

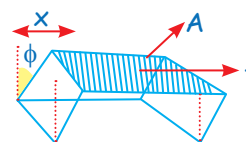
Let us consider a cube whose lower face is fixed and a tangential force F acts on the upper face whose area is A.

$$\therefore \text{Tangential stress} = F/A$$

Let the vertical sides of the cube shift through an angle ϕ , called shear strain.

Here, $E = \eta$ Modulus of Rigidity (Shear Modulus)

It is defined as the ratio of the tangential stress to the shear strain.

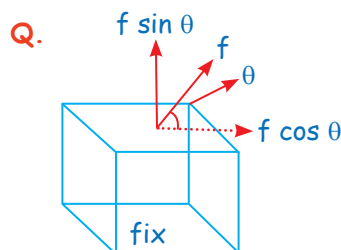


$$\text{Shear strain} = \frac{x}{L} = \tan \phi$$

$$E = \frac{F/A}{\tan \phi} = \eta$$

Where η is modulus of rigidity

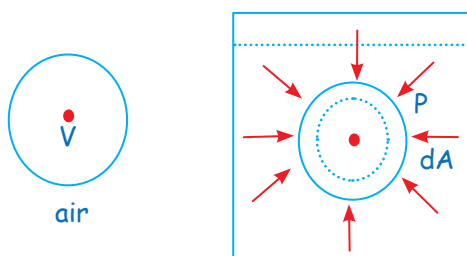
$$\eta = \frac{\text{Tangential stress}}{\text{Shear strain}} \quad \text{or} \quad \eta = \frac{F/A}{\phi} = \frac{F_{\text{tangential}}}{A\phi}$$



Sol. Shear stress = $\frac{F \cos \theta}{A}$

Longitudinal Stress = $\frac{F \sin \theta}{A}$

Bulk Stress: When a body is immersed in a water completely it surrounded by water molecule. Since the fluid presses inward on all sides of the object (figure), the solid is compressed and its volume is reduced. The fluid pressure (P) is the force per unit surface area and can be thought of as the volume stress on the solid object.



$$dF = PdA$$

$$\frac{dF}{dA} = P \text{ (Excess Pressure)}$$

For small area dA if force applied by water molecule due to pressure difference this force per unit area defined as Bulk Stress or volumetric stress.

$$\text{Vol}^n \text{ strain} = \frac{\Delta V}{V}$$

Bulk Modulus (Here, $E = B$)

$$B = \frac{\text{Vol}^n \text{ Stress}}{\text{Vol}^n \text{ Strain}}$$

$$B = \frac{-P}{\frac{\Delta V}{V}} = \frac{-\Delta P}{\Delta V / V} = \frac{-dP}{dV / V}$$

$\Delta P \rightarrow$ Excess pressure

$$\text{Compressibility} = \frac{1}{B}$$

Very Important Formula for Exam

Use 5-10 baar rough copy par likh kar practice kar lena, pata chale tu exam mein kahe



★ Isothermal bulk modulus of elasticity of gas

$$B = P \text{ (pressure of gas)}$$

★ Adiabatic bulk modulus of elasticity of gas

$$B = \gamma \times P \quad \text{where } \gamma = \frac{C_p}{C_v}$$

Q. The bulk modulus of rubber is $9.8 \times 10^8 \text{ N/m}^2$.

To what depth (approximately) a rubber ball be taken in a lake so that its volume is decreased by 0.1%?

Sol. $B = \frac{P}{\Delta V / V}$ or $P = B \frac{\Delta V}{V}$

$$\therefore P = 9.8 \times 10^8 \times \left(\frac{0.1}{100} \right)$$

$$\text{or } h\rho g = 9.8 \times 10^8 \times \left(\frac{0.1}{100} \right)$$

$$\therefore h = \frac{9.8 \times 10^8}{1000 \times 9.8} \times \frac{0.1}{100} \text{ m} = 100 \text{ m}$$

Q. A cube of length 0.01 m has its upper face displaced by 0.001 m by applying a force of 5 N tangent to surface. Find modulus of rigidity.

Sol. Shear stress = $\frac{f}{A} = \frac{5}{(0.01)^2} = 5 \times 10^4$

$$\text{Shear strain} = \frac{x}{L} = \frac{0.001}{0.01} = 0.1$$

Modulus of rigidity:

$$\frac{\text{Shear Stress}}{\text{shear Strain}} = \frac{5 \times 10^4}{0.1} = 5 \times 10^5$$

POISSON'S RATIO



$$\text{Poisson ratio } (\mu) = -\frac{\Delta r / r}{\Delta l / l} = -\frac{\Delta d / d}{\Delta l / l}$$

Relation Among Different Moduli

$$(1) Y = 3B(1 - 2\mu) \quad (2) Y = 2\eta(1 + \mu)$$

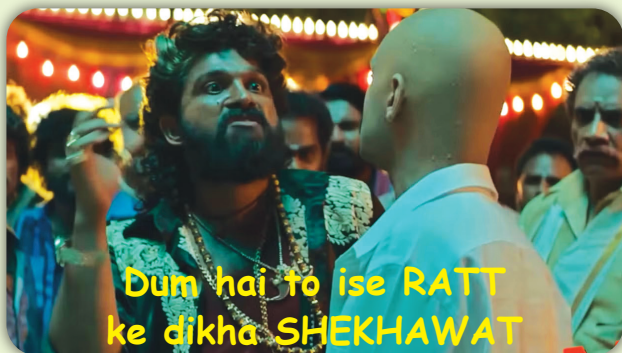
$$(3) Y = \frac{9B\eta}{3B + \eta} \quad (4) \mu = \frac{3B - 2\eta}{6B + 2\eta}$$

Y = Young's modulus

B = Bulk modulus

η = Modulus of rigidity

μ = Poisson's ratio.



Dum hai to ise RATT
ke dikha SHEKHAWAT

THERMAL STRESS AND THERMAL STRAIN

If a rod is fixed between two rigid supports, due to change in temperature its length will try to change itself and so it will exert a normal stress. Stress will be tensile if temperature decreases and compressive if temperature increases.

Change in length due to change in temperature (ΔL),
(ΔL) = $\alpha L \Delta T$, α = coefficient of thermal expansion

$$\text{Thermal strain } (E_T) = \frac{\Delta L}{L}$$

$$E_T = \alpha \Delta T$$

Thermal stress (σ_T) = Young's Modulus $\times$ Thermal strain

$$\sigma_T = Y \alpha \Delta T$$

Q. A steel wire of length 20 cm and uniform cross-section 1 mm^2 is tied rigidly at both the ends. The temperature of the wire is altered from 40°C to 20°C . Coefficient of linear expansion for steel $\alpha = 1.1 \times 10^{-5}/^\circ\text{C}$ and Y for steel is $2.0 \times 10^{11} \text{ N/m}^2$. The change in tension of the wire is:

Sol. $L = 20 \text{ cm}$

$$A = 1 \text{ mm}^2; \Delta T = 20^\circ\text{C}$$

$$\alpha = 1.1 \times 10^{-5}/^\circ\text{C}; Y = 2 \times 10^{11} \text{ N/m}^2$$

$$T = \sigma \times A = Y \alpha \Delta T \times A$$

$$= 2 \times 10^{11} \times 1.1 \times 10^{-5} \times 20 \times 10^{-6} = 44 \text{ N}$$

Q. Two rods of different metals having the same area of cross-section A are placed between the two massive walls as shown in figure. The first rod has a length l_1 , coefficient of linear expansion α_1 and Young's modulus Y_1 . The corresponding quantities for second rod are l_2 , α_2 and Y_2 . The temperature of both the rods is now raised by $t^\circ\text{C}$.



- Find the force with which the rods act on each other at higher temperature.
- Find the lengths of the rods at higher temperature.

Sol. (a) Let Δl be the displacement of the joint towards right.

$$\text{Strain on first rod} = \frac{l_1 \alpha_1 t - \Delta l}{l_1}$$

or force exerted by the first rod on the joint is

$$F_1 = Y_1 A \left(\frac{l_1 \alpha_1 t - \Delta l}{l_1} \right) \quad \dots(i)$$

$$\text{Strain on second rod} = \frac{l_2 \alpha_2 t + \Delta l}{l_2}$$

or force exerted by second rod on the joint is

$$F_2 = Y_2 A \left(\frac{l_2 \alpha_2 t + \Delta l}{l_2} \right) \quad \dots(ii)$$

In equilibrium $F_1 = F_2$

Solving this, we get

$$\Delta l = \frac{l_1 l_2 t (Y_1 \alpha_1 - Y_2 \alpha_2)}{(Y_1 l_2 + Y_2 l_1)} \quad \dots(iii)$$

Solving (i), (ii) and (iii), we get

$$F_1 = F_2 = \frac{A t (l_1 \alpha_1 + l_2 \alpha_2)}{\left(\frac{l_1}{Y_1} + \frac{l_2}{Y_2} \right)}$$

$$(b) l'_1 = l_1 + \Delta l = l_1 + \frac{l_1 l_2 t (Y_1 \alpha_1 - Y_2 \alpha_2)}{(Y_1 l_2 + Y_2 l_1)}$$

$$\text{and } l'_2 = l_2 - \Delta l = l_2 - \frac{l_1 l_2 t (Y_1 \alpha_1 - Y_2 \alpha_2)}{Y_1 l_2 + Y_2 l_1}$$

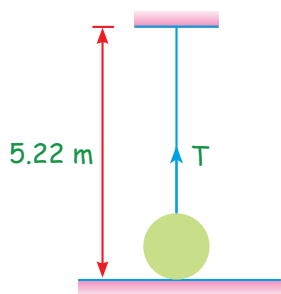
Q. A sphere of radius 0.1 m and mass 8π kg is attached to the lower end of a steel wire of length 5.0 m and diameter 10^{-3} m. The wire is suspended from 5.22 m high ceiling of a room. When the sphere is made to swing as a simple pendulum, it just grazes the floor at its lowest point. Calculate the velocity of the sphere at the lowest position. Young's modulus of steel is 1.994×10^{11} N/m².

Sol. Let Δl be the extension of wire when the sphere is at mean position. Then, we have

$$l + \Delta l + 2r = 5.22$$

$$\text{or } \Delta l = 5.22 - l - 2r = 5.22 - 5 - (2 \times 0.1) = 0.02 \text{ m}$$

Let T be the tension in the wire at mean position during oscillations, then



$$Y = \frac{T/A}{\Delta l/l}$$

$$\therefore T = \frac{YA\Delta l}{l} = \frac{Y\pi r^2 \Delta l}{l}$$

Substituting the values, we have

$$T = \frac{(1.994 \times 10^{11}) \times \pi \times (0.5 \times 10^{-3})^2 \times 0.02}{5}$$

$$= 626.43 \text{ N}$$

The equation of motion at mean position is,

$$T - mg = \frac{mv^2}{R} \quad \dots (i)$$

$$\text{Here, } R = 5.22 - r = 5.22 - 0.1 = 5.12 \text{ m}$$

$$\text{and } m = 8\pi \text{ kg} = 25.13 \text{ kg}$$

Substituting the proper values in Eq. (i), we have

$$(626.43) - (25.13 \times 9.8) = \frac{(25.13)v^2}{5.12}$$

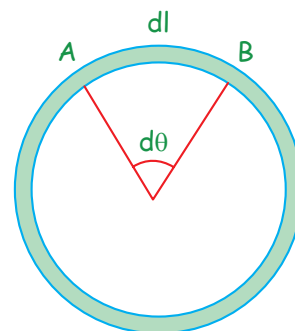
Solving this equation, we get

$$v = 8.8 \text{ m/s}$$

Q. A thin ring of radius R is made of a material of density ρ in Young's modulus Y . The ring is rotated about its centre and its own plane with angular velocity ω . Find the small increase in its radius.

Sol. Consider an element of length dl and mass dm of the ring.

Let S be the cross-sectional area of wire and T be the tension in the ring.



$$dl = R \cdot d\theta$$

$$F = 2T \sin\left(\frac{d\theta}{2}\right) = T(d\theta) \quad \left(\sin \frac{d\theta}{2} \approx \frac{d\theta}{2}\right)$$

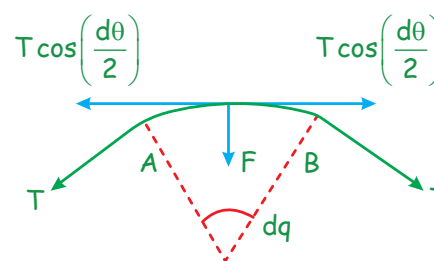
As $d\theta$ is small

This force F provides the necessary centripetal force.

$$\text{Hence, } T \cdot d\theta = (dm)R\omega^2$$

$$= (R \cdot d\theta \cdot S \rho)R\omega^2 \text{ or } \frac{T}{S} = \rho\omega^2 R^2$$

Let ΔR be the increase in radius. Then



$$\text{longitudinal strain} = \frac{\Delta L}{L} = \frac{\Delta(2\pi R)}{2\pi R} = \frac{\Delta R}{R}$$

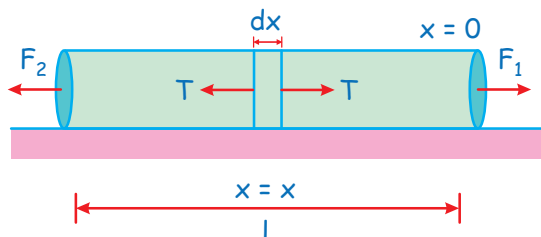
$$\text{Now } Y = \frac{\text{stress}}{\text{strain}} = \frac{T/S}{\Delta R/R}$$

$$Y = \frac{\rho\omega^2 R^3}{\Delta R}$$

$$\text{Hence } \Delta R = \frac{\rho\omega^2 R^3}{Y}$$

Q. Two opposite forces F_1 and F_2 ($< F_1$) act on an elastic plank of modulus of elasticity Y and length l placed over a smooth horizontal surface. The cross-sectional area of the plank is S . Find the change in length of the plank in the direction of the force.

Sol. Tension in the plank varies linearly from F_1 to F_2 . Therefore, tension in the plank at a distance x from the front edge is:



Change in length in element dx is:

$$dl = \frac{Tdx}{SY} \left(\Delta l = \frac{Fl}{AY} \right)$$

$$dl = \frac{\left\{ F_1 - (F_1 - F_2) \frac{x}{l} \right\}}{SY} dx$$

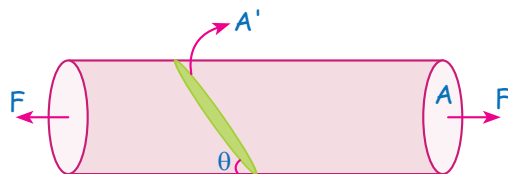
$\therefore$ Total change in length will be

$$\Delta l = \int_{x=0}^{x=l} dl$$

$$\text{or } \Delta l = \int_0^l \left\{ \frac{F_1 - (F_1 - F_2) \frac{x}{l}}{SY} \right\} dx$$

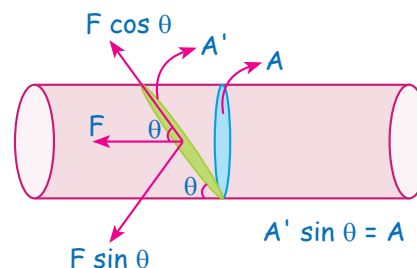
$$\Delta l = \frac{(F_1 + F_2)l}{2SY}$$

Q. If a long steel bar under a tensile stress due to forces $\vec{F}$ acting at the edges along the length of the bar. Consider a plane making an angle θ with the length. What are the tensile and shearing stresses on this plane?



- (a) For what angle is the tensile stress a maximum?
(b) For what angle is the shearing stress a maximum?

Sol.



$$\text{Tensile Stress} = \frac{F \sin \theta}{A'} = \frac{F \sin \theta}{\frac{A}{\sin \theta}} = \frac{F \sin^2 \theta}{A}$$

$$\text{Shearing Stress} = \frac{F \cos \theta}{A'} = \frac{F \cos \theta}{\frac{A}{\sin \theta}} = \frac{F \sin 2\theta}{2A}$$

$$\text{Tensile Stress}_{\max} = \frac{F}{A}; \left(\theta = \frac{\pi}{2} \right)$$

$$\text{Shearing Stress}_{\max} = \frac{F}{2A}; \left(\theta = \frac{\pi}{4} \right)$$

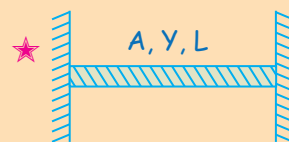


Kam ka Dabba

dekh bhai is chapter ke 80 to 90% questions niche jo formula likh rha hu us par based milenge

$$\star \frac{F}{A} = Y \frac{\Delta l}{l}$$

$$\star \Delta l = \frac{Fl}{AY}$$



$$\Delta l = \frac{Tl}{Ay} = l \alpha \Delta \theta$$

On increasing temp. by $\Delta \theta$

$$\star B = -\frac{dP}{dV/V}$$

$$\star \text{Poisson ratio} = -\frac{\Delta r/r}{\Delta l/l}$$